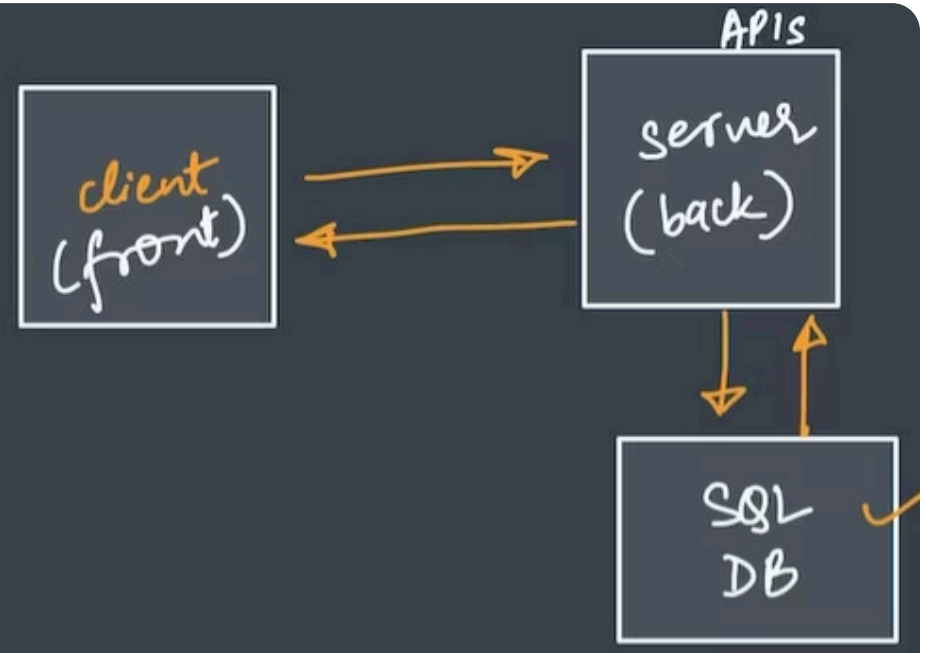
Node with SQL

Connecting the power of Node.js with the reliability of SQL databases to build fast, scalable, and data-driven applications.

MySQL2 Package

To connect Node with MySQL

```
connection.end(); // to close connection
```



https://www.npmjs.com/package/mysql2

https://www.npmjs.com/package/@faker-js/faker

SQL command line

```
:\Users\NithinShivanna>mysql -u root -p
nter password: **********
elcome to the MySQL monitor.  Commands end with ; or \g.
our MySQL connection id is 15
erver version: 8.0.31 MySQL Community Server - GPL

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```

CREATE Table user

```
CREATE TABLE user (
 id ~~userId~~ VARCHAR(50) PRIMARY KEY,
    username VARCHAR(50) UNIQUE,
    email VARCHAR(50) UNIQUE NOT NULL,
    password VARCHAR(50) NOT NULL
);
```

id	username	email	password

INSERT User

using placeholders

```
let user = ["123@abc2", "random_user2", "random@gmail.com2", "random@123"];

connection.query(
  `INSERT INTO user (userId, username, email, password) VALUES (?, ?, ?, ?)`,
  user,
  function (err, results) {
    if (err) throw err;
    console.log(results);
  }
);
```

```javascript
// A simple SELECT query
connection.query(
  'SELECT * FROM `table` WHERE `name` = "Page" AND `age` > 45',
  function (err, results, fields) {
    console.log(results); // results contains rows returned by server
    console.log(fields); // fields contains extra meta data about results, if available
  }
);

// Using placeholders
connection.query(
  'SELECT * FROM `table` WHERE `name` = ? AND `age` > ?',
  ['Page', 45],
  function (err, results) {
    console.log(results);

    // Close the connection
    connection.end();
  }
);
```

```javascript
//Inserting New Data
let q = "INSERT INTO user (id, username, email, password) VALUES (?, ?, ?, ?)";
let user = ["123b", "123_newuserb", "abc@gmail.comb", "abcb"];

try {
  connection.query(q, user, (err, result) => {
    if (err) throw err;
    console.log(result);
  });
} catch (err) {
  console.log(err);
}
```

```
let q = "INSERT INTO user (id, username, email, password) VALUES ?";
let users = [
  ["123b", "123_newuserb", "abc@gmail.comb", "abcb"],
  ["123c", "123_newuserc", "abc@gmail.comc", "abcc"],
];

try {
  connection.query(q, [users], (err, result) => {
    if (err) throw err;
    console.log(result);
  });
} catch (err) {
  console.log(err);
}
```

INSERT in Bulk

using faker

```
let data = [];
for (let i = 1; i <= 100; i++) {
  data.push(getUser());
}

console.log(data);
let q = `INSERT INTO user (userId, username, email, password) VALUES ?`;

connection.query(q, [data], function (err, results) {
  if (err) throw err;
  console.log(results);
});
```

```
let getUser = () => {
  return [
    faker.datatype.uuid(),
    faker.internet.userName(),
    faker.internet.email(),
    faker.internet.password(),
  ];
};
```

Do Activity

Routing

REST

GET / → show no of user in DB

GET /user → show users (email, id, username)

PATCH /user/:id → username edit

GET / Fetch & show total number of users on our app

```
app.get("/", (req, res) => {
  let q = `SELECT count(*) FROM user`;
  try {
    connection.query(q, (err, result) => {
      if (err) throw err;
      let count = result[0]["count(*)"];
      res.render("home.ejs", { count });
    });
  } catch (err) {
    res.send("some error occurred");
  }
});
```

GET /user Fetch & show (userId, username, email) of all users

```
app.get("/user", (req, res) => {
  let q = `SELECT * FROM user`;
  try {
    connection.query(q, (err, result) => {
      if (err) throw err;
      let data = result;
      console.log(data);
      res.render("users.ejs", { data });
    });
  } catch (err) {
    res.send("some error occurred");
  }
});
```

Creating Our Routes EDIT

GET /user/:id/edit	To get form to edit the username, based on id This form will require a password
PATCH /user/:id	To edit username, if correct password was entered in form

Homework Tasks

1) Create Form to Add a new user to the Database /user POST

2) Create Form to Delete a user from Database if they enter correct email id & password.